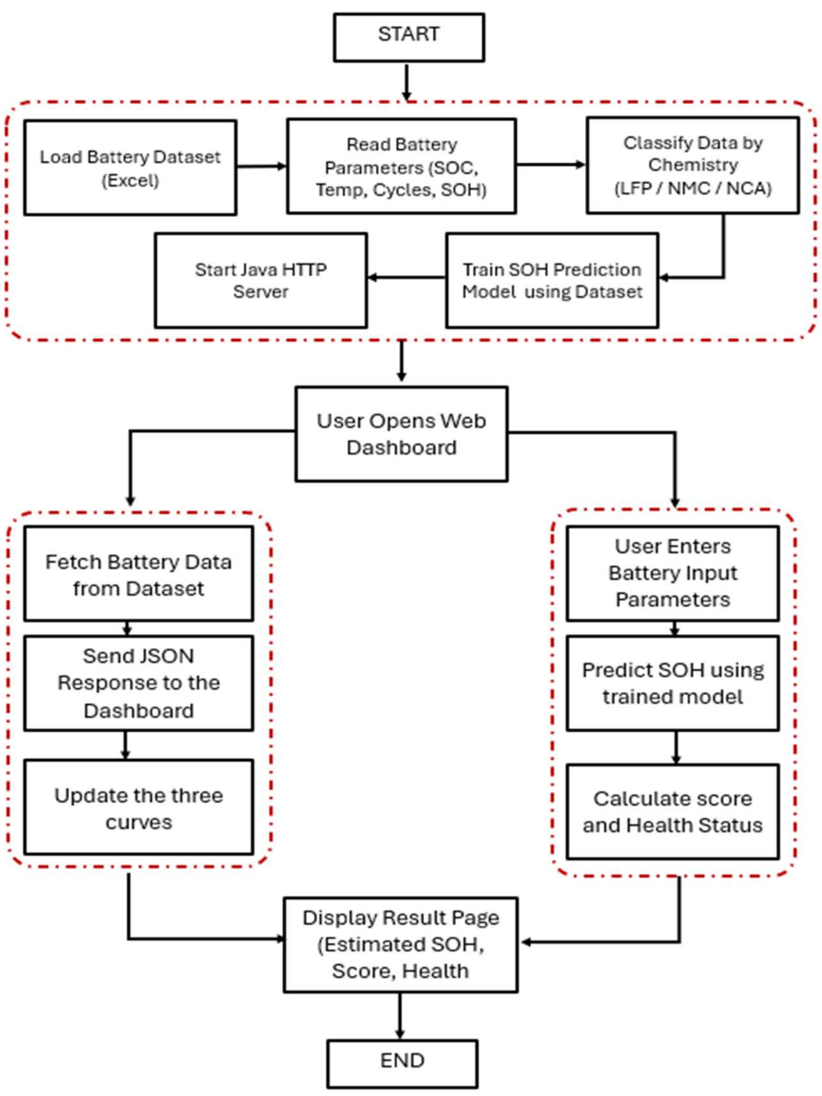
EV BATTERY HEALTH MONITORING AND VISUALIZATION SYSTEM

Student Name: S.MADAN (23BEE1006) and K. KARTHIK (23BEE1015)

Research Focus:

This work focuses on developing a dataset with various parameters such as (SOC, Temperature, Clock, etc..) to monitor and analyze the Real-time Battery of an Electric Vehicle with various chemistries so as to choose the Best or Optimal Battery. The various Parameters that are being measured are used to calculate a final weightage score, State of Health(SOH). This score provides meaningful insight into the Battery's Life (Good, Warning or End of Life). The final score is also presented as a user-friendly real-time dashboard enabling effective visualization and comparison. This research aims to support Predictive Maintenance, Improve Battery Reliability, and assist stakeholders in making appropriate decisions regarding Battery Usage and Management.

Flow chart:

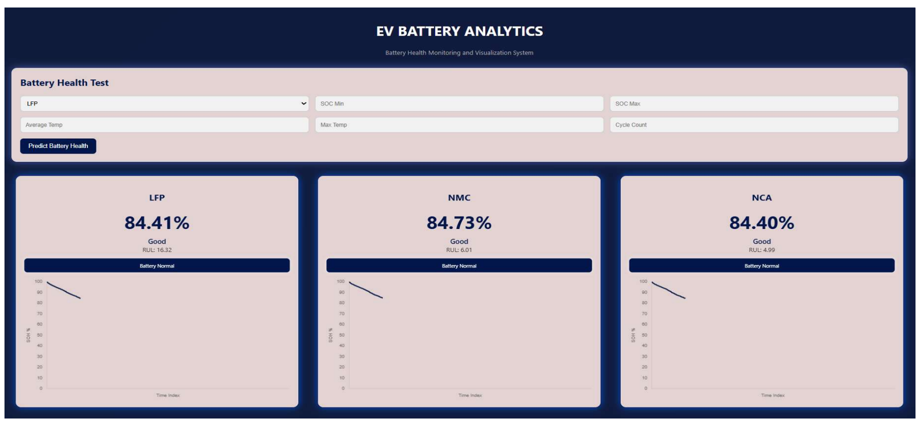


Specifications of the Batteries (LFP, NMC and NCA)

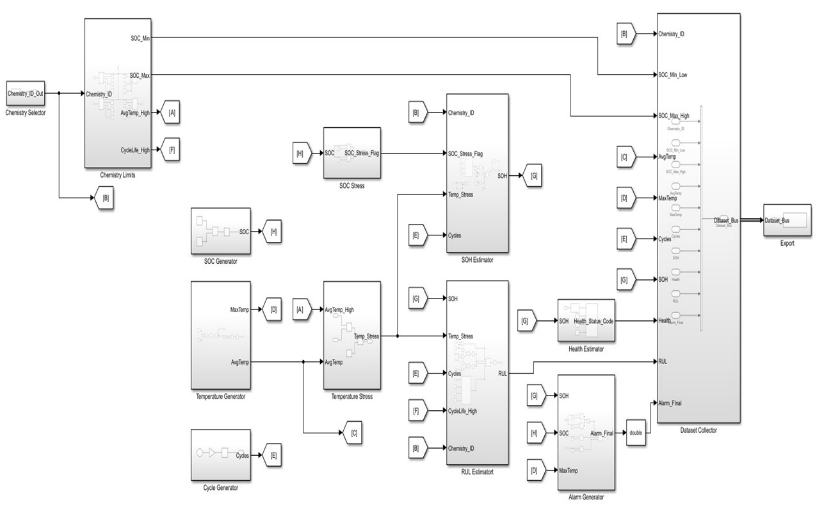
PARAMETERS	LFP	NMC	NCA
Chemistry ID	1	2	3
SOC Minimum	10–20%	20%	20%
SOC Maximum	90–100%	80–90%	80–90%
Average Temperature	10–35 °C	15–35 °C	20–35 °C
Maximum Temperature	≤45–50 °C	≤45–50 °C	≤45–50 °C
Cycles	3000–7000+	1000–2000	500–1500

Dashboard of the Website:

The dashboard of my project displays the performance of the battery for various chemistries such as LFP, NMC and NCA. This helps the user to calculate the State of Health(SOH), Remaining Useful Life(RUL) and Battery Status using Graphs and Alarm Systems.



Dataset Generation using MATLAB:



Conclusion:

This project successfully develops a smart system for monitoring and predicting the Battery Health using Simulation and Analysis. This project provides real-time insights about the Battery Performance, degradation and remaining life which in turn helps in improving the Battery Efficiency and Supporting Reliable Operations.